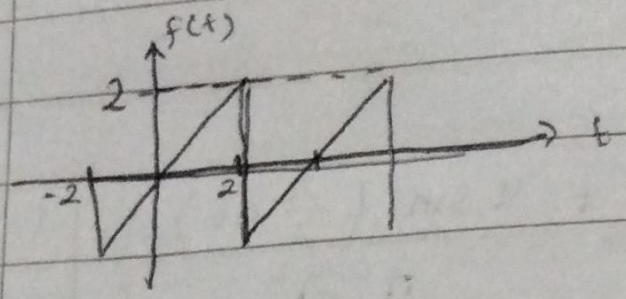
Tugas Fourier 2

8. fungsi $f(t)$ seperti ditunjukkan pada gambar berikut!



$T = 4$ $\quad -2 \le t \le 2$

$f(t) = t$ $\quad \omega = \frac{2\pi}{T} = \frac{2\pi}{4} = \frac{\pi}{2}$

$f(t)$ fungsi ganjil karena $f(-t) = -f(t)$

$f(-t) = -t$

$a_0 = 0$

$a_n = 0$

$$b_n = \frac{4}{T}\int_0^{T/2} f(t)\sin(n\omega t)\,dt$$

$$= \frac{4}{4}\int_0^{2} t\sin\left(n\tfrac{\pi}{2}t\right)dt$$

$$= \left(\frac{\sin\left(n\frac{\pi}{2}t\right)}{\left(n\frac{\pi}{2}\right)^2} - \frac{t\cos\left(n\frac{\pi}{2}t\right)}{\left(n\frac{\pi}{2}\right)^2}\right)\Bigg|_0^2 = \frac{\sin n\pi}{\left(\frac{n\pi}{2}\right)^2} - \frac{2\cos(n\pi)}{\left(\frac{n\pi}{2}\right)^2}$$

$\sin n\pi = 0$, $\cos n\pi \begin{cases} -1, & n \text{ ganjil} \\ +1, & n \text{ genap} \end{cases}$ maka:

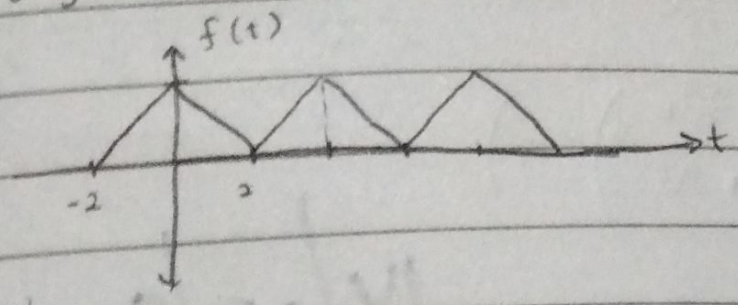
$$b_n = \frac{-2\cos n\pi}{\left(\frac{n\pi}{2}\right)^2} \begin{cases} b_n = \frac{8}{n^2\pi^2}, & n \text{ ganjil} \\ b_n = \frac{-8}{n^2\pi^2}, & n \text{ genap} \end{cases}$$

Deret fourier : $f(t) = \frac{8}{\pi^2}\left(\sin\frac{\pi}{2}t - \frac{1}{4}\sin\frac{2\pi}{2}t + \frac{1}{9}\sin\frac{3\pi}{2}t + \cdots\right)$

No.

Date:

10 Fungsi $f(t)$ seperti ditunjukkan pada gambar berikut!



$T = 4$ $\quad f(t) \begin{cases} \frac{t}{2}+1, & -2 \le t \le 0 \\ -\frac{t}{2}+1, & 0 \le t \le 2 \end{cases}$

$\omega = \frac{2\pi}{4} = \frac{\pi}{2}$

$f(t)$ adalah fungsi genap karena simetris sb y.

$$a_0 = \frac{2}{T}\int_0^{T/2} f(t)\,dt$$

$$= \frac{2}{4}\int_0^2 \frac{-t}{2}+1\,dt = \frac{1}{2}\left(\frac{-t^2}{4}+t\right)\Big|_0^2 = \frac{1}{2}$$

$$a_n = \frac{4}{T}\int_0^{T/2} f(t)\cos(n\omega t)\,dt$$

$$= \frac{4}{4}\int_0^2 \left(\frac{-t}{2}+1\right)\cos\frac{n\pi}{2}t$$

$\int u\,dv = uv - \int v\,du$

misal $u = -\frac{t}{2}+1 \rightarrow du = -\frac{1}{2}$

$dv = \cos\left(\frac{n\pi}{2}t\right) \rightarrow v = \frac{2\sin\left(\frac{n\pi}{2}t\right)}{n\pi}$, maka:

$$a_n = \left(\frac{-t}{2}+1\right)\left(\frac{2\sin\left(\frac{n\pi}{2}t\right)}{n\pi}\right) - \int \frac{2\sin\left(\frac{n\pi}{2}t\right)}{n\pi}\cdot\left(\frac{-1}{2}\right)dt$$

$$= \frac{-2\cos n\pi}{n^2\pi^2} - \left(\frac{-2}{n^2\pi^2}\right) = \frac{2-\cos n\pi}{n^2\pi^2}$$

$\cos n\pi \begin{cases} -1, & n \text{ ganjil} \\ 1, & n \text{ genap} \end{cases}$ maka:

$$a_n = \begin{cases} \frac{3}{n^2\pi^2}, & n \text{ ganjil} \\ \frac{1}{n^2\pi^2}, & n \text{ genap} \end{cases}$$

$b_n = 0$

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Dered fourier : $f(t) = \frac{1}{2} + \left(\frac{3}{\pi^2} \cos \frac{\pi}{2} t + \frac{1}{2^2 \pi^2} \cos \frac{2\pi}{2} t + \frac{3 \cos \frac{3\pi}{2} t}{3^2 \pi^2} + \right.$

$\left. \frac{\cos \frac{4\pi}{2} t}{4^2 \pi^2} + \dots \right)$

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